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UNIT II

Data Warehouse Design and OLAP

2.1 Three-Tier Data Warehouse Architecture

A data warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data used to support management decision-making. Most data warehouses are organized into three tiers, separating data storage from data access so each layer can be optimized and scaled independently.

The Three Tiers

Tier	Also Called	Function
Bottom Tier	Data Warehouse Server	Stores the actual warehouse data, typically on a relational database server; this is where ETL loads cleansed and integrated data from source systems
Middle Tier	OLAP Server	Provides a multidimensional view of the data, implemented via ROLAP, MOLAP, or HOLAP engines, sitting between the storage layer and the front-end tools
Top Tier	Front-End / Client Tier	Contains the query, reporting, analysis, and data-mining tools that end users interact with directly

Data Flow Through the Architecture

```
Source Systems (OLTP DBs, files, external feeds)
       |
       |   ETL (Extract, Transform, Load)
       v
BOTTOM TIER  --  Data Warehouse Server (relational DB, star/snowflake)
       |
       |   OLAP engine (ROLAP / MOLAP / HOLAP)
       v
MIDDLE TIER  --  OLAP Server (multidimensional cube view)
       |
       |   Query & analysis requests
       v
TOP TIER  --  Front-End Tools (reporting, dashboards, ad-hoc query)
```

OLAP Server Types

1. ROLAP (Relational OLAP) — stores data in relational tables and generates SQL dynamically to compute multidimensional views on the fly
2. MOLAP (Multidimensional OLAP) — pre-computes and stores data in a proprietary multidimensional cube structure for very fast retrieval
3. HOLAP (Hybrid OLAP) — combines both: detailed data stays relational (ROLAP-style) while summary data is pre-aggregated into cubes (MOLAP-style)

2.2 Enterprise Data Warehouse, Data Mart, and Operational Data Store (ODS)

Organizations rarely build a single monolithic warehouse. Three related but distinct constructs are commonly used together:

Construct	Scope	Typical Use
Enterprise Data Warehouse (EDW)	Organization-wide; integrates data from all business functions/departments	Cross-departmental strategic reporting and analysis
Data Mart	Subset of the EDW, scoped to one department or business line (e.g. Sales, HR)	Faster, department-specific reporting without querying the full warehouse
Operational Data Store (ODS)	Current, near-real-time operational data, lightly integrated	Operational reporting; often acts as a staging point before data is loaded into the EDW

Note: An ODS is NOT a data warehouse: it holds current, frequently-updated (volatile) data for short-term operational reporting, whereas a warehouse holds historical, non-volatile, time-variant data for long-term analysis.

Dependent vs Independent Data Marts

- Dependent Data Mart — sourced FROM the central EDW (top-down approach; ensures consistency across marts)
- Independent Data Mart — built directly from source systems without a central EDW (bottom-up approach; faster to build, but risks inconsistent definitions across marts)

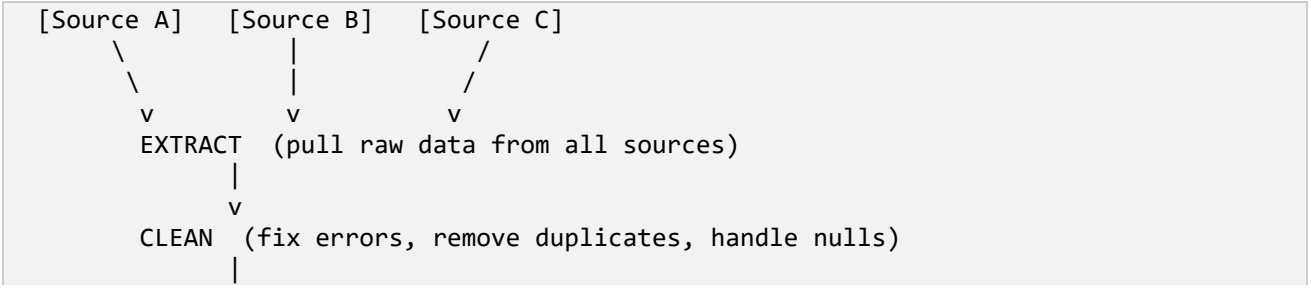
2.3 ETL (Extract, Transform, Load) Process

ETL is the set of processes used to move data from heterogeneous source systems into the data warehouse in a clean, consistent, and query-ready form. It typically runs as a scheduled batch process, though real-time/near-real-time variants (sometimes called ELT or streaming ETL) also exist.

The Four Stages

Stage	What Happens
Data Extraction	Data is pulled from source systems — relational databases, flat files, APIs, legacy systems — often using full extraction (entire dataset) or incremental extraction (only changed records since the last run)
Data Cleaning	Errors, inconsistencies, duplicates, and missing values in the extracted data are detected and corrected
Data Transformation	Cleaned data is converted into the warehouse's target format — includes standardizing units, aggregating, joining, and deriving new attributes
Data Loading	Transformed data is written into the warehouse's target tables — via full load (initial bulk load) or incremental load (ongoing periodic updates)

ETL Pipeline



```
          v
    TRANSFORM  (standardize formats, aggregate, derive fields)
          |
          v
    LOAD  (write into warehouse fact/dimension tables)
          |
          v
    [Data Warehouse]
```

Common Data Cleaning Tasks

- Handling missing values — via deletion, mean/median substitution, or predictive imputation
- Removing duplicate records
- Correcting inconsistent naming/formatting (e.g. “NY” vs “New York”)
- Identifying and handling outliers
- Resolving referential integrity issues (orphaned foreign keys)

2.4 Metadata Management

Metadata is “data about data” — it describes the structure, source, meaning, and lineage of the data stored in the warehouse. Without well-managed metadata, a warehouse becomes a “data swamp” that is difficult to trust or navigate.

Types of Metadata in a Data Warehouse

Type	Describes	Example
Technical Metadata	Physical structure and technical properties of data	Table/column names, data types, keys, ETL job schedules
Business Metadata	Business meaning and context of the data	Business definitions of terms like “Active Customer”, ownership, report descriptions
Operational Metadata	Runtime and process information	ETL job run times, row counts loaded, error logs, data lineage

2.5 Multidimensional Data Model — Facts and Dimensions

Unlike the normalized (2D relational) model used in OLTP systems, data warehouses use a multidimensional model, conceptually visualized as a data cube, optimized for analytical queries (“slice this by that”) rather than transaction processing.

Facts vs Dimensions

Aspect	Fact Table	Dimension Table
Contains	Quantitative, measurable business metrics (e.g. sales amount, quantity sold)	Descriptive, textual attributes that give context to facts (e.g. product name, customer city)
Size	Very large — one row per business event/transaction	Comparatively small — one row per distinct entity (product, customer, date)
Keys	Composed mainly of foreign keys referencing dimension tables, plus the measures	A primary key (often a surrogate key) uniquely identifying each dimension member

Aspect	Fact Table	Dimension Table
Example	Sales_Fact(date_key, product_key, store_key, quantity, revenue)	Product_Dim(product_key, product_name, category, brand)

Types of Measures in a Fact Table

- Additive — can be summed across all dimensions (e.g. sales revenue, units sold)
- Semi-additive — can be summed across some dimensions but not others (e.g. account balance can be summed across accounts, but not meaningfully across time)
- Non-additive — cannot be meaningfully summed at all (e.g. a ratio like profit margin percentage)

2.6 Star, Snowflake, and Fact Constellation Schemas

The multidimensional model is physically implemented in a relational database using one of three common schema designs.

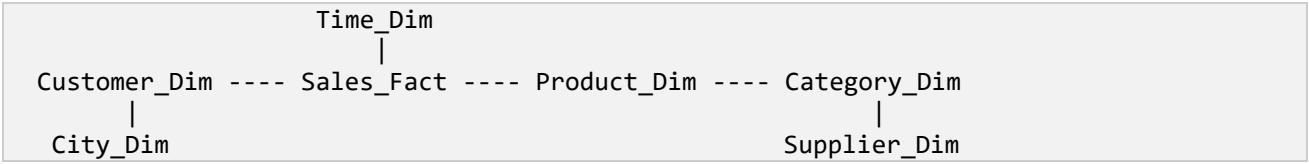
Star Schema

A star schema has a single, central fact table connected directly to multiple denormalized dimension tables, resembling a star shape. Dimension tables are NOT normalized — some redundancy is accepted in exchange for simpler, faster queries (fewer joins).



Snowflake Schema

A snowflake schema extends the star schema by normalizing dimension tables into multiple related sub-dimension tables, reducing data redundancy at the cost of requiring more joins for queries — resembling a snowflake shape.



Fact Constellation Schema (Galaxy Schema)

A fact constellation schema contains multiple fact tables that share one or more common dimension tables, modeling several related business processes at once (e.g. Sales and Shipping sharing the same Time and Product dimensions).

Schema Comparison

Aspect	Star Schema	Snowflake Schema	Fact Constellation
Fact Tables	One	One	Multiple
Dimension Normalization	Denormalized	Normalized	Denormalized or normalized
Query Joins	Fewer, simpler	More, complex	Complex (multiple fact tables)
Redundancy	Higher	Lower	Varies

Aspect	Star Schema	Snowflake Schema	Fact Constellation
Query Performance	Faster	Slower (more joins)	Depends on query

Building a Simple Star Schema — Steps

1. Identify the business process to model (e.g. retail sales)
2. Determine the grain — the level of detail one fact row represents (e.g. one row per line-item per transaction)
3. Identify the dimensions relevant to the process (e.g. Time, Product, Store, Customer)
4. Identify the facts/measures to be recorded (e.g. quantity, unit price, total amount)

2.7 OLAP Concepts and OLAP Operations

OLAP (Online Analytical Processing) refers to a category of tools and techniques for interactively analyzing multidimensional data from multiple perspectives. It contrasts with OLTP (Online Transaction Processing), which is optimized for high-volume, simple read/write transactions rather than complex analytical queries.

OLTP vs OLAP

Aspect	OLTP	OLAP
Purpose	Day-to-day transaction processing	Analysis and decision support
Data	Current, detailed, frequently updated	Historical, summarized, mostly read-only
Schema	Highly normalized (3NF)	Denormalized (star/snowflake)
Query Type	Simple, short transactions	Complex, multidimensional, aggregate queries
Users	Clerks, front-line staff	Analysts, managers, executives

Core OLAP Operations

OLAP tools let users navigate a data cube interactively using the following standard operations:

Operation	What It Does
Roll-up	Aggregates data by climbing UP a concept hierarchy (e.g. from City to State to Country) or by reducing the number of dimensions — summarizing detail into higher-level totals
Drill-down	The reverse of roll-up: navigates DOWN a concept hierarchy to see more detailed data (e.g. from Year to Quarter to Month)
Slice	Selects a single value for one dimension, producing a sub-cube (e.g. viewing sales for only Year = 2025, across all products and regions)
Dice	Selects specific value ranges for TWO OR MORE dimensions, producing a smaller sub-cube (e.g. Year in {2024, 2025} AND Region = “West”)
Pivot (Rotate)	Rotates the data axes to provide an alternative presentation — e.g. swapping rows and columns to view Product vs Region instead of Region vs Product

Visualizing Roll-up and Drill-down

```
Concept Hierarchy (Location):   Country -> State -> City
```

```
ROLL-UP  (City -> Country)          DRILL-DOWN  (Country -> City)
-------------------------           -------------------------
City-level detail                    Country-level total
       |                                     |
       v  (aggregate up)                     v  (expand down)
Country-level total                  City-level detail
```

Slice and Dice on a 3D Cube

```
Cube dimensions: [Time] x [Product] x [Region]

SLICE:  Time = "2025"  ->  a 2D slice: [Product] x [Region] for 2025 only

DICE:   Time in {2024,2025} AND Region = "West"
        ->  a smaller cube: [Time subset] x [Product] x [Region=West]
```

OLAP Operations — Practice Walkthrough

1. Start with a full sales cube across Time, Product, and Region
2. Apply Slice: fix Region = “North” to get a 2D Time × Product view
3. Apply Drill-down on Time: expand “Q1 2025” into Jan, Feb, Mar
4. Apply Pivot: swap the axes so Product becomes rows and Time becomes columns
5. Apply Roll-up on Product: aggregate individual products up into their Product Category

2.8 Unit II — Summary & Practice Questions

Key Takeaways

- The three-tier architecture separates storage (bottom), analytical processing (middle/OLAP server), and presentation (top/front-end tools)
- An EDW is organization-wide; a Data Mart is department-scoped; an ODS holds current operational data and is not itself a warehouse
- ETL moves data through Extraction, Cleaning, Transformation, and Loading stages to populate the warehouse
- Metadata management (technical, business, operational) is essential for a warehouse to remain trustworthy and navigable
- Fact tables hold measures; dimension tables hold descriptive context; together they form the multidimensional model
- Star schema favors query speed via denormalization; snowflake schema favors reduced redundancy via normalization; fact constellation models multiple related processes
- OLAP's five core operations — roll-up, drill-down, slice, dice, and pivot — let analysts navigate a data cube from different angles

Practice Questions

1. Describe the three-tier data warehouse architecture and explain the role of each tier.
2. Differentiate between an Enterprise Data Warehouse, a Data Mart, and an Operational Data Store.
3. Explain the four stages of the ETL process with a suitable diagram.
4. What is the difference between a fact table and a dimension table? Give an example of each.
5. Compare the star schema and snowflake schema. When would you prefer one over the other?
6. What is a fact constellation schema? How does it differ from a star schema?
7. Explain the OLAP operations roll-up, drill-down, slice, dice, and pivot with examples.
8. Differentiate between OLTP and OLAP systems on at least four parameters.

UNIT III

Introduction to Data Mining and Data Preprocessing

3.1 Introduction to Data Mining

Data mining is the process of discovering interesting, non-trivial, previously unknown, and potentially useful patterns, relationships, or knowledge from large volumes of data. It draws on techniques from statistics, machine learning, database systems, and pattern recognition.

> **Note:** Data mining is often used loosely as a synonym for the broader Knowledge Discovery in Databases (KDD) process, but strictly speaking it refers to just one step within KDD — the step where patterns are actually extracted.

Why Data Mining?

- Explosive growth of data collected by organizations (transactions, web logs, sensors, social media)
- Data alone is not knowledge — raw data must be analyzed to reveal actionable patterns
- Manual analysis does not scale to the volume and dimensionality of modern datasets

3.2 Knowledge Discovery in Databases (KDD)

KDD is the overall, multi-step process of discovering useful knowledge from data, of which data mining is the core analytical step. The process is typically iterative — insights from later steps often send you back to refine an earlier step.

Steps in the KDD Process

1. Data Cleaning — remove noise, inconsistent data, and handle missing values
2. Data Integration — combine data from multiple, possibly heterogeneous, sources
3. Data Selection — retrieve only the data relevant to the analysis task from the integrated data
4. Data Transformation — consolidate data into forms appropriate for mining (e.g. aggregation, normalization)
5. Data Mining — apply intelligent methods to extract data patterns (the core analytical step)
6. Pattern Evaluation — identify the patterns that are genuinely interesting, based on some measure of interestingness
7. Knowledge Presentation — visualize and present the mined knowledge to the user, e.g. through reports, charts, or dashboards

KDD Process Flow

```
Raw Data
   |  1. Cleaning
   v
Clean Data
   |  2. Integration
   v
Integrated Data
   |  3. Selection
   v
Task-Relevant Data
   |  4. Transformation
   v
Transformed Data
   |  5. Data Mining
   v
Patterns
```

```
    |  6. Pattern Evaluation
    v
Interesting Patterns
    |  7. Knowledge Presentation
    v
Knowledge  (delivered to the user)
```

3.3 Data Mining Process and Architecture

Beyond KDD's conceptual steps, a practical data mining system has a concrete architecture of interacting components.

Major Components of a Data Mining Architecture

Component	Role
Database / Data Warehouse Server	Fetches the relevant data based on the user's data mining request
Knowledge Base	Domain knowledge used to guide the search or evaluate the interestingness of resulting patterns
Data Mining Engine	The core module; performs the actual mining functions (association, classification, clustering, etc.)
Pattern Evaluation Module	Interacts with the mining engine to focus the search toward interesting patterns, using interestingness measures
Graphical User Interface (GUI)	Enables communication between the user and the data mining system, and visualizes results

A Typical Data Mining Process (Applied)

1. Business/Problem Understanding — define the goal of the mining exercise
2. Data Understanding — collect and explore initial data to identify quality issues
3. Data Preparation — clean, integrate, transform, and reduce the data (see Section 3.6)
4. Modeling — apply data mining techniques/algorithms (classification, clustering, association rules, etc.)
5. Evaluation — assess whether the model/patterns actually meet the business objective
6. Deployment — put the resulting knowledge/model into operational use

3.4 Functionalities and Issues in Data Mining

Major Functionalities

Functionality	Description	Example
Class/Concept Description	Summarizes data belonging to a class using characterization or discrimination	Summarizing the characteristics of “High-value customers”
Association Rule Mining	Discovers relationships/co-occurrences between items in large datasets	“Customers who buy bread also buy butter” (market basket analysis)
Classification	Predicts a categorical class label for new data based on a learned model	Classifying an email as “spam” or “not spam”

Functionality	Description	Example
Regression / Prediction	Predicts a continuous numeric value	Predicting next month's sales revenue
Clustering	Groups similar data objects together without predefined class labels	Segmenting customers into behavior-based groups
Outlier / Anomaly Analysis	Identifies data that deviates significantly from the norm	Detecting fraudulent credit card transactions
Evolution / Trend Analysis	Describes patterns of objects whose behavior changes over time	Analyzing stock price trends over several years

Major Issues in Data Mining

- Mining Methodology Issues — handling different kinds of knowledge, data types, and the need for domain-specific mining
- User Interaction Issues — interactive mining at multiple abstraction levels, incorporating background/domain knowledge, presenting results usefully
- Performance Issues — efficiency and scalability of mining algorithms on very large datasets; parallel and distributed mining
- Diverse Data Type Issues — mining structured, semi-structured, and unstructured data (text, multimedia, spatial, web data)
- Data Privacy and Security Issues — protecting individual privacy while still extracting useful aggregate patterns

3.5 Types of Data and Data Quality

Types of Data Mined

Data Type	Description	Example
Structured Data	Highly organized, fits a fixed schema (rows/columns)	Relational database tables
Semi-Structured Data	Has some organizational structure but not a rigid schema	XML/JSON files, emails with headers
Unstructured Data	No predefined structure	Free text, images, audio, video
Time-Series / Sequence Data	Data recorded with an explicit temporal or sequential order	Stock prices, sensor readings, DNA sequences
Spatial Data	Data with a geographic or spatial component	GPS coordinates, maps, geospatial imagery

Dimensions of Data Quality

Dimension	Question It Answers
Accuracy	Does the recorded value correctly reflect the real-world value?
Completeness	Are all required values present, with no missing attributes?
Consistency	Are related data values in agreement across the dataset/sources?

Dimension	Question It Answers
Timeliness	Is the data up to date at the point it is used?
Believability	How much is the data trusted by users, based on its source and history?
Interpretability	How easily can the data be understood by the intended user?

3.6 Data Preprocessing Techniques

Real-world data is typically incomplete, noisy, and inconsistent. Data preprocessing transforms raw data into a clean, well-structured form suitable for effective mining. It is often the most time-consuming stage of the entire KDD process.

Typical Preprocessing Order

1. Data Cleaning — fill in missing values, smooth noisy data, identify/remove outliers, resolve inconsistencies
2. Data Integration — merge data from multiple sources, resolving schema conflicts and redundancy (e.g. the same entity recorded differently across sources)
3. Data Transformation — normalize, aggregate, and generalize data into forms suitable for mining
4. Data Reduction — reduce data volume while preserving analytical integrity, via dimensionality reduction, numerosity reduction, or compression
5. Data Discretization — convert continuous attributes into discrete intervals or categories (often treated as a special case of data reduction)

Data Cleaning — Common Techniques

Problem	Common Techniques
Missing Values	Ignore the record; fill manually; use a global constant; use attribute mean/median; use the most probable value (via regression or decision-tree prediction)
Noisy Data	Binning (smoothing by bin means/medians/boundaries); regression (fit data to a function); clustering (detect and remove outliers)
Inconsistent Data	Use metadata/domain knowledge and manual or semi-automated correction routines

Data Integration Challenges

- Entity Identification Problem — matching equivalent real-world entities from different sources (e.g. “cust_id” vs “customer_number”)
- Redundancy — the same attribute may have different names across sources, or be derivable from other attributes; correlation analysis can detect this
- Detection and Resolution of Data Value Conflicts — the same real-world entity may have attribute values that differ across sources (e.g. different units of measurement)

Data Transformation — Common Strategies

Strategy	Description
Normalization	Scales attribute values to fall within a small, specified range (e.g. min-max normalization to [0,1], or z-score normalization)
Aggregation	Summarizes data — e.g. daily sales aggregated into monthly sales totals

Strategy	Description
Generalization	Replaces low-level (raw) data with higher-level concepts using concept hierarchies (e.g. exact street address generalized to city)
Attribute Construction	Constructs new attributes from existing ones to help the mining process (e.g. deriving "BMI" from height and weight)

Data Reduction Strategies

Strategy	Description
Dimensionality Reduction	Reduces the number of attributes/variables under consideration, e.g. via Principal Component Analysis (PCA) or attribute subset selection
Numerosity Reduction	Replaces the original data volume with a smaller representation, e.g. via regression models, histograms, clustering, or sampling
Data Compression	Applies transformations to obtain a reduced or compressed representation of the original data (lossy or lossless)

Data Discretization Methods

1. Binning — divides continuous values into a fixed number of bins (equal-width or equal-frequency)
2. Histogram Analysis — partitions values into buckets based on their frequency distribution
3. Cluster Analysis — groups nearby continuous values into the same discrete cluster/category
4. Concept Hierarchy Generation — recursively splits or merges values into higher-level concepts (e.g. exact age → age group → life stage)

Feature Selection Methods

Feature selection is a specific technique, closely related to dimensionality reduction, that selects a relevant subset of the original attributes rather than transforming them into new ones.

1. Filter Methods — select features based on statistical measures (e.g. correlation, chi-square, information gain) independent of any mining algorithm
2. Wrapper Methods — evaluate feature subsets by actually running a mining algorithm on them and measuring resulting performance
3. Embedded Methods — perform feature selection as an intrinsic part of the model-building process itself (e.g. decision tree attribute selection, LASSO regression)

3.7 Applications of Data Mining

Domain	Typical Application
Retail / Market Basket Analysis	Association rule mining to discover which products are frequently purchased together
Banking and Finance	Credit risk scoring, fraud detection, customer segmentation
Healthcare	Disease prediction and diagnosis support, patient risk stratification, drug interaction discovery
Telecommunications	Customer churn prediction, network fault detection
Web / Social Media	Recommendation systems, sentiment analysis, community/link analysis

Domain	Typical Application
Manufacturing	Predictive maintenance, quality control, process optimization
Education	Predicting student performance, personalizing learning paths (educational data mining)

3.8 Unit III — Summary & Practice Questions

Key Takeaways

- Data mining is the pattern-extraction step within the broader KDD process, which spans cleaning through knowledge presentation
- A data mining architecture combines a data source, knowledge base, mining engine, pattern evaluation module, and a user-facing GUI
- Core data mining functionalities include classification, association, clustering, regression, and outlier/trend analysis
- Data comes in structured, semi-structured, unstructured, time-series, and spatial forms, and its quality is judged on accuracy, completeness, consistency, timeliness, believability, and interpretability
- Preprocessing proceeds through cleaning, integration, transformation, reduction, and discretization to prepare raw data for mining
- Feature selection (filter, wrapper, embedded methods) picks a relevant attribute subset, distinct from dimensionality reduction techniques that transform attributes

Practice Questions

1. Explain the steps involved in the Knowledge Discovery in Databases (KDD) process.
2. Describe the major components of a typical data mining system architecture.
3. List and briefly explain any five functionalities of data mining, with an example of each.
4. What are the major issues in data mining? Discuss any three in detail.
5. Explain the six dimensions of data quality with suitable examples.
6. What is data preprocessing? Explain data cleaning and data transformation with examples.
7. Differentiate between dimensionality reduction and feature selection.
8. Explain any four real-world applications of data mining.